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Bearing unit

Abstract

A bearing unit (**10**) has a radially outer ring (**31**), a radially inner ring (**33**), an eccentric locking collar (**40**), a row of rolling bodies (**32**), and a sealing device (**50**). The sealing device (**50**) is arranged on the same side as the locking collar (**40**) and is interposed between the radially inner ring (**33**), the radially outer ring (**31**) and the locking collar (**40**). The sealing device (**50**) includes a metal shield (**55**) stably connected to the radially outer ring (**31**), and an elastomeric component (**60**) co-molded on the shield (**55**) in a radially inward direction. The elastomeric component (**60**) has a plurality of lips, of which at least one lip (**64, 65**) is in contact with the locking collar (**40**).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Italian Application No. 102023000002691, filed Feb. 16, 2023, the entirety of which is hereby incorporated by reference.

FIELD

(2) The present disclosure relates to a bearing unit. The bearing unit is suitable for use in the manufacturing sector, in particular in the agricultural and mining sectors, and for other demanding applications characterized by dusty environments.

BACKGROUND

(3) The bearing unit generally has a first component, for example a radially outer ring, which is fastened to a stationary element, usually a containment casing, and a second component, for example a radially inner ring, which is fastened to a rotary element, normally a rotary shaft. In other applications, the radially inner ring may be stationary and the radially outer ring may be rotary. In any case, the rotation of one ring in relation to the other in the roller bearing units is enabled by a plurality of rolling bodies that are positioned between the two surfaces of the components normally referred to as raceways. The rolling bodies may be balls, cylindrical or conical rollers, needle rollers, and similar rolling bodies.

(4) It is known for bearing units to have sealing devices to protect the raceways and rolling bodies against external contaminants and to seal the lubricating grease. Typically, the sealing devices are made up of a co-moulded elastomer gasket on a shaped metal shield mounted by interference or chamfering in a seat in the rings of the bearing unit, for example the radially outer ring. The gasket is provided with at least one contacting or non-contacting sealing lip that forms a seal by means of sliding contact with the other ring of the bearing unit, or by means of a labyrinth formed with the same ring.

(5) Unlike other applications, the bearing units used in agricultural applications often only have this type of sealing device and do not have any other protection against external contaminants (such as a second axially outer metal shield). This is related to the common risk in agricultural applications of plant fibres coming between the second shield and the gasket, removing the shield from the bearing unit, and exposing the gaskets directly to contaminants.

(6) Consequently, the sealing devices described above are used on their own in agricultural applications, and performance is improved by changing the number of contacting lips. However, this solution inevitably increases axial size, which is somewhat incompatible with the systems for fastening the radially inner ring to the rotary shaft. This problem is particularly relevant in applications that use an eccentric locking collar as a fastening system.

SUMMARY

(7) The present disclosure is therefore intended to provide a bearing unit that does not have the drawbacks described above. This is achieved by a novel sealing device that uses the locking system as part of the sealing system, so as to create a direct contact between the locking collar and the sealing lips of the sealing device.

(8) The present disclosure provides a bearing unit having the features set out in the attached claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is described below with reference to the attached drawings, which show some non-limiting example embodiments of the bearing unit, in which:

(2) FIG. **1** is a cross-sectional view of a preferred embodiment of the bearing unit of the present disclosure, and

(3) FIG. **2** is a magnified cross-sectional view of a detail of the bearing unit in FIG. **1**.

DETAILED DESCRIPTION

(4) In FIG. **1**, reference sign **10** denotes a bearing unit as a whole that is intended mainly, but not exclusively, for use in the agricultural sector. The bearing unit **10** may comprise a stationary radially outer ring **31**. The bearing unit **10** may comprise a radially inner ring **33** that is rotary about a central axis X of rotation of the bearing unit **10**, and constrained to rotate with a rotary shaft **5** by an eccentric locking collar **40**. The bearing unit **10** may comprise a row of rolling bodies **32**, in this case balls, interposed between the radially outer ring **31** and the radially inner ring **33** to enable relative rotation. The bearing unit **10** may comprise a cage **34** for the rolling bodies to hold the

rolling bodies of the row of rolling bodies **32** in position.
(5) Throughout the present description and in the claims, terms and expressions indicating position and orientation, such as “radial” and “axial”, should be understood with reference to the axis X of rotation of the bearing unit **10**.
(6) The bearing unit **10** is also provided with two sealing devices arranged axially on opposite sides of the bearing unit **10**, to seal said unit from the external environment. In particular, a first sealing device is arranged on the side opposite the locking collar **40**, and is of a known type. For this reason, the sealing device **35** is shown only schematically in the figure and is not further described.
(7) A second sealing device **50** is arranged on the same side as the locking collar **40** and is the subject of the present disclosure. Indeed, according to the present disclosure, the sealing device is provided with a suitable number of lips to create a contact with the eccentric locking collar **40**.
(8) Advantageously, the sealing device is shaped to create a labyrinth with the locking collar **40**.
(9) The compatibility of the sealing device with the locking collar advantageously helps the entire sealing system to enhance protection of the bearing unit. In other words, the present disclosure creates a synergy between the closing system and the sealing system.
(10) With reference to FIG. **2**, the second sealing device **50** (hereinafter referred to more simply as the sealing device **50**) is interposed between the radially inner ring **33**, the radially outer ring **31** and the locking collar **40**. In particular, the sealing device **50** is stably fastened in a radially inner slot **31***a* of the radially outer ring **31** and, by means of the contacting lips thereof, creates a seal with the radially inner ring **33** and the locking collar **40**, as described in greater detail below.
(11) The sealing device **50** may comprise a metal shield **55** stably connected to the radially outer ring **31**. The sealing device **50** may comprise an elastomeric component **60** co-molded on the shield **55** in a radially inward direction, having an approximately frustoconical base structure **60′**, and provided with a plurality of lips, of which at least one lip **64**, **65** is in contact with the locking collar **40**.
(12) Preferably, as shown in the embodiment in FIG. **2**, both of the lips **64**, **65** are in contact with the locking collar **40**.
(13) The metal shield **55** may be provided with a radially outer flange portion **56** having a V-shaped cross section. The flange portion **56** has a radially outer rounded vertex **57** stably inserted into the seat **31***a* of the radially outer ring **31**. The metal shield **55** may be provided with a radially inner, axially outer appendage **58**. The metal shield **55** may be provided with a frustoconical central portion **59** that connects the flange portion **56** to the appendage **58**.
(14) As already mentioned, the elastomeric component **60** is in turn provided with a plurality of lips, and in particular may be provided with a radially and axially inner radial protuberance **61** intended to contain the grease placed inside the bearing unit. The radial protuberance **61** is not in contact with a radially outer surface **33***a* of the radially inner ring **33**. The elastomeric component **60** may be provided with a first lip **62** that is radially inside and axially outside the radial protuberance **61**, the first lip **62** being in contact with the surface **33***a* of the radially inner ring **33**. The elastomeric component **60** may be provided with a second lip **63** that is radially inside and axially outside the first lip **62**, the second lip **63** also being in contact with the surface **33***a* of the radially inner ring **33** by means of a first contact area **631** and a second contact area **632**. The elastomeric component **60** may be provided with a third lip **64** in the shape of a “V” opening radially outward in cross section. The axially outer third lip **64** is radially outside the aforementioned lips. It is in contact with an axially inner first surface **40***a* of the locking collar **40**. It is formed by two mutually adjacent portions: a root **641** that is connected to the base **60′** of the elastomeric component **60**, and a distal portion **642** that, under the effect of the axial compression caused by the axial contact with the locking collar **40**, is folded over the root **641** to define the aforementioned “V” shape. The distal portion **642** of the third lip **64** has a contacting tip **643** delimited by two surfaces: a first surface **644** approximately transverse to the axis X and a second surface **645** approximately parallel to the axis X. The first surface **644** is in contact with the surface

40*a* of the locking collar **40**. The elasticity provided by the "V" shape makes the contact as uniform as possible, thereby improving sealing performance. The elastomeric component **60** may be provided with an axially outer fourth lip **65** radially outside the third lip **64**, the fourth lip **65** also being in contact with the axially inner surface **40***a* of the locking collar **40**. The fourth lip **65** extends radially from the base **60′** of the elastomeric component **60** towards the locking collar **40** and has a contacting tip **651** delimited by two surfaces: a first surface **652** approximately transverse to the axis X and a second surface **653** approximately parallel to the axis X. The first surface **652** is in contact with the surface **40***a* of the locking collar **40**. The tip **651** of the fourth lip **65** and the tip **643** of the third lip **64** are arranged at a predetermined radial distance from one another.

(15) An annular loop **60″** is defined between the base **60′** of the elastomeric component **60**, the third lip **64** and the fourth lip **65**. In use, the annular loop **60″** is completely isolated from the outside when the tip **643** of the third lip **64** and the tip **651** of the fourth lip **65** come into contact with the surface **40***a* of the locking collar **40**. This creates a chamber C between the third lip **64**, the fourth lip **65** and the first surface **40***a* of the locking collar **40**. The external contaminants are stopped in this chamber C and create an obstacle for contaminants that could subsequently enter the sealing device **50**.

(16) Advantageously, the chamber C may also be filled with fresh grease before assembling the bearing unit **10**. This creates a contact zone between the sealing device **50** (in particular the third lip **64**) and the locking collar **40**, said contact zone also being used to lubricate the locking collar itself. Simultaneously, the presence of grease, which is known to have high viscosity, creates a very viscous, swamp-like barrier against the ingress of external contaminants.

(17) Advantageously, the elastomeric component **60** may be provided with an axially outer axial protuberance **66** radially outside the fourth lip **65**. The axial protuberance **66** forms a labyrinth L with a radially outer second surface **40***b* of the locking collar **40**.

(18) The seals created by the third lip **64** and by the fourth lip **65**, both of which are in contact with the locking collar **40**, and the seal created by the labyrinth L between the axial protuberance **66** and the locking collar **40** all improve the sealing system as a whole, which, as already mentioned, also benefits from cooperation of the locking collar **40**.

(19) The labyrinth L, or the gap between the axial protuberance **66** and the locking collar **40**, can indeed be considered to be a further seal, protecting all of the axially more internal lips of the sealing device **50** against external contaminants, and consequently improving the protection of the bearing unit **10**.

(20) The radial dimension of the labyrinth L is preferably between 0.2 mm and 0.4 mm. Any contact between the axial protuberance **66** and the locking collar **40** would not create a problem other than a slight increase in friction torque, but would also simultaneously improve the protection of the bearing unit **10** against external contaminants. This solution works better in all applications in which the friction torque is not very high (agricultural sector, heavy industry, mining, etc.).

(21) To create the labyrinth L, the shield **55**, and in particular the central portion **59** thereof, need simply be designed so that the elastomeric component **60** (in particular the axial protuberance **66** thereof) axially overlaps the locking collar **40** (in particular the second surface **40***b* thereof). In short, it is primarily the axial length of the central portion **59** of the shield **55** that should be adjusted.

(22) Another important effect of this dimensioning is that the shield **55** and the elastomeric component **60** with its axial protuberance **66** completely overlap, in all cases in the axial direction, all of the lips of the elastomeric component **60**, creating an "umbrella" effect that protects the lips from the external environment.

(23) The axial dimension of the radially inner ring **33** guaranteeing contact between the lips of the elastomeric component **60** and the locking collar **40** is smaller than in known solutions. This reduces the axial dimensions of the bearing unit **10**, which benefits end users.

(24) Reducing the axial dimension of the radially inner ring **33**, during assembly, the locking collar

40 comes into contact with and bends the third lip **64** and the fourth lip **65** upon contact between a metal component (the collar) and an elastomeric component.
(25) In particular, the third lip **64** is the lip that alone provides the sealing device **50** with a high performance seal, and is highly flexible as a result of its “V” shape. This makes it possible to provide relative axial positioning between the sealing device **50** and the locking collar **40** so that the interference between the third lip **64** and the first surface **40***a* is between 0.5 mm and 0.7 mm, these values being greater than the interference values for contacting lips in known sealing devices (interference values not greater than 0.4 mm).
(26) Again with the objective of enhancing the performance of the sealing device **50**, the fourth lip **65** is a slightly contacting lip arranged axially inside the labyrinth L, i.e. immediately downstream of the labyrinth L along the path of a potential contaminant entering from the external environment towards the bearing unit. The interference of the fourth lip **65** with the first surface **40***a* of the locking collar **40** should not be less than 0.1 mm, even under the worst possible tolerance conditions. The lesser interference for this fourth lip **65** compared to the interference provided for the third lip **64** can be explained by the need to avoid an excessively adverse effect on friction torque. In any case, even with such interference values, the fourth lip **65** is nonetheless useful in protecting the bearing unit **10** and the grease contained therein against lightweight external contaminants such as dust and small particles.
(27) In short, the sealing device according to the present disclosure improves the protection of the bearing unit against external contaminants, and consequently the service life of said bearing unit.
(28) Numerous other variants exist in addition to the embodiments of the present disclosure described above. These embodiments should also be understood to be examples and do not limit the scope, applications or possible configurations of the present disclosure. Indeed, although the description provided above enables the person skilled in the art to carry out the present disclosure at least according to one example configuration thereof, numerous variations of the components described could be used without thereby departing from the scope of the present disclosure, as defined in the attached claims interpreted literally and/or according to their legal equivalents.

Claims

1. A bearing unit comprising: a stationary radially outer ring, a radially inner ring rotatable with respect to an axis of rotation, an eccentric locking collar configured to constrain the radially inner ring to rotation with a rotating shaft, a row of rolling bodies interposed between the radially outer ring and the radially inner ring, and a sealing device arranged on the same side as the locking collar, the sealing device being interposed between the radially outer ring, the radially inner ring and the locking collar, the sealing device comprising: a shield made of metallic material, the shield stably connected to the radially outer ring, and an elastomeric component co-molded on the shield in a radially internal direction, the elastomeric component having a plurality of lips, wherein at least one lip of the plurality of lips is in contact with the locking collar.
2. The bearing unit according to claim 1, wherein the elastomeric component comprises: a radial protuberance, radially and axially internal, which has the function of containing the grease located inside the bearing unit, a first lip, radially internal and axially external with respect to the radial protuberance and contacting a surface of the radially inner ring, a second lip, radially internal and axially external with respect to the first lip and also contacting the surface of the radially inner ring, a third lip, V-shaped, radially external with respect to the first and second lips and axially external, the third lip being in contact with a first surface, axially internal, of the locking collar, and a fourth lip, radially external with respect to the third lip and axially external, the fourth lip also being in contact with the surface of the locking collar.
3. The bearing unit according to claim 2, wherein the elastomeric component is provided with an axial protuberance, radially external with respect to the fourth lip and axially external, the axial

protuberance cooperating with a second surface, radially external, of the locking collar to define a labyrinth.
4. The bearing unit according to claim 3, wherein a radial dimension of the labyrinth is between 0.2 mm and 0.4 mm.
5. The bearing unit according to claim 2, wherein the interference, in use, between the third lip and the first surface of the locking collar is between 0.5 mm and 0.7 mm.
6. The bearing unit according to claim 2, wherein the interference, in use, between the fourth lip and the first surface of the locking collar is not less than 0.1 mm.
7. The bearing unit according to claim 2, wherein the third lip, fourth lip and first surface of the locking collar define an accumulation chamber for external contaminants.
8. The bearing unit according to claim 7, wherein the chamber contains lubricating grease.
9. The bearing unit according to claim 8, wherein the shield comprises: a flange portion, radially external, V-shaped, an appendage, radially internal and axially external, and a central portion, with a truncated cone shape, which connects the flange portion to the appendage.
10. The bearing unit according to claim 1, wherein the shield comprises: a flange portion, radially external, V-shaped, an appendage, radially internal and axially external, and a central portion, with a truncated cone shape, which connects the flange portion to the appendage.
11. The bearing unit according to claim 10, wherein the flange portion has a rounded vertex, radially external, permanently inserted into a seat of the radially outer ring.
